Case Study-Based Database Design and Implementation

Programme Name & Branch : B.Tech / CSE

Course Name & Code **:** BCSE302P / Database Systems Lab

Class Number **:** VL20262701033465

Slot **:** L11+L12

- Read the assigned case study carefully and understand the business requirements before starting the database design.
- Students must provide screenshots of successful execution and outputs for important SQL commands and queries.
- Students must submit the complete executable SQL script along with the project report.
- The project report must be technically accurate, properly formatted, and free from unnecessary duplication.
- Students must be prepared to demonstrate the project and explain their database design, SQL queries, constraints, relationships, and outputs during evaluation.
- Individual contribution must be clearly identifiable. Every student should be capable of explaining the complete project.
- Copying database designs, SQL scripts, reports, or outputs from other groups or online sources is strictly prohibited. Any similarity identified during evaluation may result in reduction of marks.
- The project will be evaluated primarily on database design quality, correctness, SQL implementation, integration of concepts, analytical capability, originality, and the student's ability to justify the design decisions.

Sl. No	Evaluation Component	Marks
1	**Database Design** (Case Study Analysis, ER Diagram, Entities, Relationships and Relational Schema)	10
2	**Database Implementation** (DDL, DML and Constraints)	5
3	**SQL Query Development**	10
4	**Project Report and Demonstration**	7
5	**Viva**	8
	Total	**40**

- The Project Presentation will be conducted from **26.10.2026 to 30.10.2026**.
- Project Report should be uploaded in Assessment 10.

CASE STUDY 1: Blood Bank Management System

A regional blood bank plans to develop a Blood Donation, Inventory and Distribution Management System to replace its existing manual and spreadsheet-based blood inventory process. The proposed system should manage information related to donors, blood groups, donations, blood units, storage facilities, hospitals, patients, blood requests, compatibility information, and blood issue transactions. It should support donor management, blood inventory monitoring, hospital requests, blood-unit allocation, expiry monitoring, and generation of analytical reports for blood-bank administrators. The system must ensure data consistency, integrity, reduced redundancy, accurate blood inventory management, and efficient retrieval of blood availability information. Analyse the given case study and independently design, develop, and evaluate a relational database solution for the proposed system. The entities, attributes, relationships, table names, and relational schema must be identified and designed by the student based on the requirements of the case study.

1. Requirement Analysis and Database Design

Analyse the case study and:

- Identify the functional requirements and business rules.
- Identify appropriate entities and attributes.
- Identify relationships and cardinalities.
- Design a suitable ER diagram.
- Convert the ER model into a relational schema.
- Identify appropriate primary keys and foreign keys.

2. Database Implementation and Data Management

Implement the proposed database using SQL.

The implementation should demonstrate:

- Appropriate database structure creation and modification.
- Suitable integrity constraints.
- Insertion of realistic donor, donation, blood-unit, hospital, and request data.
- Modification and deletion of records.
- Enforcement of data consistency and referential integrity.
- Appropriate management of blood-unit availability and expiry.

The database should contain sufficient interrelated data to support meaningful analysis.

3. SQL-Based Data Processing and Analysis

Develop SQL queries to generate meaningful information from the database.

The queries should address scenarios such as:

- Blood-group-wise inventory.
- Donor-wise donation history.
- Hospital-wise blood demand.
- Blood-unit issue statistics.
- Blood donation trends.
- Expiring blood units.
- Highest and lowest available blood groups.
- Hospital request statistics.
- Donor participation.
- Blood groups requiring urgent replenishment.

Appropriate techniques for data filtering, transformation, calculation, grouping, and summarization must be demonstrated.

4. Advanced SQL: Joins, Subqueries and Views

Develop advanced SQL solutions based on the requirements of the case study.

The implementation must include:

- Appropriate joins involving multiple relations.
- Subqueries for solving complex blood-bank requirements.
- At least one correlated subquery.
- At least two meaningful views for presenting derived blood-bank information.

The student must develop queries such as determining:

- Blood groups having stock below the overall average.
- Hospitals having demand above the average hospital demand.
- Donors with higher-than-average donation frequency.
- Blood groups having repeated shortages.
- Blood units approaching expiry.
- Hospitals generating repeated urgent requests.
- Blood availability by blood group.
- Other relevant management-level insights.

The choice and purpose of each advanced query must be explained and justified.

5. Blood Availability Risk Identification

Create a Blood Availability Risk Identification and Decision-Support Module using the database designed by you. The blood bank wants to identify blood groups that may experience shortages based on current inventory, historical demand, recent requests, and expiring units.

Develop an appropriate solution that analyses blood availability in relation to relevant demand benchmarks and generates a report such as:

Blood Group	Available Units	Monthly Demand	Expiring Units	Recent Requests	Risk Status
O−	18	52	4	21	CRITICAL SHORTAGE
A+	185	140	8	34	ADEQUATE

The criteria for classifying a blood group as CRITICAL SHORTAGE, LOW STOCK, or ADEQUATE must be defined and justified by the student.

The solution must integrate information from multiple relations and demonstrate effective use of advanced SQL and database-driven decision support.

CASE STUDY 2: E-Governance Citizen Service Management System

A state government plans to develop an E-Governance Citizen Service Management System to replace its existing manual and spreadsheet-based citizen-service application management process. The proposed system should manage information related to citizens, government departments, government services, service applications, supporting documents, government officials, appointments, approvals, rejections, and service-delivery timelines. It should support citizen service applications, document verification, application processing, approval management, service tracking, and generation of analytical reports for government administration. The system must ensure data consistency, integrity, reduced redundancy, transparency, and efficient retrieval of citizen-service information. Analyse the given case study and independently design, develop, and evaluate a relational database solution for the proposed system. The entities, attributes, relationships, table names, and relational schema must be identified and designed by the student based on the requirements of the case study.

1. Requirement Analysis and Database Design

Analyse the case study and:

- Identify the functional requirements and business rules.
- Identify appropriate entities and attributes.
- Identify relationships and cardinalities.
- Design a suitable ER diagram.
- Convert the ER model into a relational schema.
- Identify appropriate primary keys and foreign keys.

2. Database Implementation and Data Management

Implement the proposed database using SQL.

The implementation should demonstrate:

- Appropriate database structure creation and modification.
- Suitable integrity constraints.
- Insertion of realistic citizen, service, application, official, and approval data.
- Modification and deletion of records.
- Enforcement of data consistency and referential integrity.

The database should contain sufficient interrelated data to support meaningful analysis.

3. SQL-Based Data Processing and Analysis

Develop SQL queries to generate meaningful information from the database.

The queries should address scenarios such as:

- Department-wise application volume.
- Service-wise application statistics.
- Application approval and rejection rates.
- Average processing time.
- Officer workload.
- Pending applications.
- Delayed applications.
- Citizen service history.
- Departments requiring additional resources.
- Services having high rejection rates.

Appropriate techniques for data filtering, transformation, calculation, grouping, and summarization must be demonstrated.

4. Advanced SQL: Joins, Subqueries and Views

Develop advanced SQL solutions based on the requirements of the case study.

The implementation must include:

- Appropriate joins involving multiple relations.

- Subqueries for solving complex e-governance requirements.
- At least one correlated subquery.
- At least two meaningful views for presenting derived government-service information.

The student must develop queries such as determining:

- Departments having higher-than-average pending applications.
- Officials handling more applications than the average official.
- Services having above-average rejection rates.
- Applications taking longer than departmental average processing time.
- Citizens submitting repeated applications.
- Departments requiring additional processing capacity.
- Service-level performance.
- Other relevant administrative insights.

The choice and purpose of each advanced query must be explained and justified.

5. Government Service Delay Risk Identification

Create a Government Service Delay Risk Identification and Decision-Support Module using the database designed by you. The government wants to identify applications that may exceed their service-level deadline based on processing time, SLA deadline, department workload, document completeness, and service complexity.

Develop an appropriate solution that analyses application processing performance in relation to relevant service benchmarks and generates a report such as:

Application	Department	Processing Days	SLA Days	Officer Workload	Risk Status
APP101	Revenue	18	15	87	CRITICAL DELAY
APP205	Transport	4	15	32	ON TRACK

The criteria for classifying an application as CRITICAL DELAY, AT RISK, or ON TRACK must be defined and justified by the student.

The solution must integrate information from multiple relations and demonstrate effective use of advanced SQL and database-driven decision support.

CASE STUDY 3: Agricultural Farm and Crop Management System

An agricultural organization plans to develop an Agricultural Farm and Crop Management System to replace its existing manual and spreadsheet-based farm management process. The proposed system should manage information related to farmers, farms, soil conditions, crops, seeds, fertilizers, irrigation activities, pesticides, harvesting, crop yield, and agricultural resources. It should support farm planning, crop monitoring, resource utilization, yield analysis, and generation of analytical reports for agricultural management. The system must ensure data consistency, integrity, reduced redundancy, and efficient retrieval of agricultural information. Analyse the given case study and independently design, develop, and evaluate a relational database solution for the proposed system. The entities, attributes, relationships, table names, and relational schema must be identified and designed by the student based on the requirements of the case study.

1. Requirement Analysis and Database Design

Analyse the case study and:

- Identify the functional requirements and business rules.
- Identify appropriate entities and attributes.
- Identify relationships and cardinalities.
- Design a suitable ER diagram.
- Convert the ER model into a relational schema.
- Identify appropriate primary keys and foreign keys.

2. Database Implementation and Data Management

Implement the proposed database using SQL.

The implementation should demonstrate:

- Appropriate database structure creation and modification.
- Suitable integrity constraints.
- Insertion of realistic farm, crop, soil, irrigation, fertilizer, and harvest data.
- Modification and deletion of records.
- Enforcement of data consistency and referential integrity.

The database should contain sufficient interrelated data to support meaningful analysis.

3. SQL-Based Data Processing and Analysis

Develop SQL queries to generate meaningful information from the database.

The queries should address scenarios such as:

- Farm-wise crop yield.
- Crop-wise production.
- Farmer productivity.
- Fertilizer consumption.
- Irrigation frequency.
- Pest incidents.
- Crop profitability.
- Highest and lowest yielding farms.
- Soil-condition statistics.
- Farms requiring agricultural intervention.

Appropriate techniques for data filtering, transformation, calculation, grouping, and summarization must be demonstrated.

4. Advanced SQL: Joins, Subqueries and Views

Develop advanced SQL solutions based on the requirements of the case study.

The implementation must include:

- Appropriate joins involving multiple relations.
- Subqueries for solving complex agricultural analysis requirements.

- At least one correlated subquery.
- At least two meaningful views for presenting derived agricultural information.

The student must develop queries such as determining:

- Farms producing above or below crop averages.
- Farmers having above-average productivity.
- Crops having lower-than-average yield.
- Farms with repeated pest incidents.
- Farms having unusually high resource consumption.
- Crops requiring agricultural intervention.
- Farm-level productivity.
- Other relevant management-level insights.

The choice and purpose of each advanced query must be explained and justified.

5. Crop Yield Risk Identification

Create a Crop Yield Risk Identification and Decision-Support Module using the database designed by you. The agricultural organization wants to identify farms that may experience below-normal crop yield based on historical yield, soil conditions, irrigation adequacy, fertilizer utilization, and pest incidents.

Develop an appropriate solution that analyses crop performance in relation to relevant agricultural benchmarks and generates a report such as:

Farm	Crop	Current Yield	Historical Average	Pest Incidents	Irrigation Status	Risk Status
Farm A	Rice	2.8 t	4.6 t	5	Low	HIGH RISK
Farm B	Wheat	5.1 t	4.8 t	1	Adequate	NORMAL

The criteria for classifying a farm as HIGH RISK, MODERATE RISK, or NORMAL must be defined and justified by the student.

The solution must integrate information from multiple relations and demonstrate effective use of advanced SQL and database-driven decision support.

CASE STUDY 4: Construction Project and Resource Management System

A construction organization plans to develop a Construction Project and Resource Management System to replace its existing spreadsheet-based project monitoring process. The proposed system should manage information related to clients, construction projects, contractors, workers, materials, suppliers, equipment, project tasks, milestones, expenses, inspections, and project progress. It should support project planning, resource allocation, expense tracking, milestone monitoring, quality inspection, and generation of analytical reports for construction management. The system must ensure data consistency, integrity, reduced redundancy, and efficient retrieval of project information. Analyse the given case study and independently design, develop, and evaluate a relational database solution for the proposed system. The entities, attributes, relationships, table names, and relational schema must be identified and designed by the student based on the requirements of the case study.

1. Requirement Analysis and Database Design

Analyse the case study and:

- Identify the functional requirements and business rules.
- Identify appropriate entities and attributes.
- Identify relationships and cardinalities.
- Design a suitable ER diagram.
- Convert the ER model into a relational schema.
- Identify appropriate primary keys and foreign keys.

2. Database Implementation and Data Management

Implement the proposed database using SQL.

The implementation should demonstrate:

- Appropriate database structure creation and modification.
- Suitable integrity constraints.
- Insertion of realistic project, contractor, worker, material, equipment, and expense data.
- Modification and deletion of records.
- Enforcement of data consistency and referential integrity.

The database should contain sufficient interrelated data to support meaningful analysis.

3. SQL-Based Data Processing and Analysis

Develop SQL queries to generate meaningful information from the database.

The queries should address scenarios such as:

- Project progress.
- Budget utilization.
- Contractor workload.
- Worker allocation.
- Material consumption.
- Equipment utilization.
- Task completion.
- Milestone performance.
- Inspection statistics.
- Project expenditure.

Appropriate techniques for data filtering, transformation, calculation, grouping, and summarization must be demonstrated.

4. Advanced SQL: Joins, Subqueries and Views

Develop advanced SQL solutions based on the requirements of the case study.

The implementation must include:

- Appropriate joins involving multiple relations.
- Subqueries for solving complex project-management requirements.

- At least one correlated subquery.
- At least two meaningful views for presenting derived project information.

The student must develop queries such as determining:

- Projects exceeding their average budget utilization.
- Contractors handling above-average workload.
- Projects having delayed milestones.
- Projects with above-average material consumption.
- Equipment having unusually high utilization.
- Projects with repeated inspection failures.
- Project-level resource performance.
- Other relevant management-level insights.

The choice and purpose of each advanced query must be explained and justified.

5. BTL-6 Challenge: Construction Project Delay Risk Identification

Create a Construction Project Delay Risk Identification and Decision-Support Module using the database designed by you. The construction organization wants to identify projects that may experience schedule delays based on milestone completion, budget utilization, pending tasks, material availability, and workforce availability. Develop an appropriate solution that analyses project progress in relation to relevant project benchmarks and generates a report such as:

Project	Completion %	Budget Used %	Pending Tasks	Material Availability	Risk Status
Project A	48	82	34	Low	HIGH DELAY RISK
Project B	76	69	8	Adequate	ON SCHEDULE

The criteria for classifying a project as HIGH DELAY RISK, MODERATE RISK, or ON SCHEDULE must be defined and justified by the student.

The solution must integrate information from multiple relations and demonstrate effective use of advanced SQL and database-driven decision support.

CASE STUDY 5: Insurance Policy and Claims Management System

An insurance organization plans to develop an Insurance Policy and Claims Management System to replace its existing spreadsheet-based policy and claims management process. The proposed system should manage information related to customers, insurance policies, policy types, agents, beneficiaries, premiums, claims, claim assessments, claim settlements, and policy renewals. It should support policy management, premium tracking, claim processing, agent performance analysis, and generation of analytical reports for insurance management. The system must ensure data consistency, integrity, reduced redundancy, accurate claim processing, and efficient retrieval of insurance information. Analyse the given case study and independently design, develop, and evaluate a relational database solution for the proposed system. The entities, attributes, relationships, table names, and relational schema must be identified and designed by the student based on the requirements of the case study.

1. Requirement Analysis and Database Design

Analyse the case study and:

- Identify the functional requirements and business rules.
- Identify appropriate entities and attributes.
- Identify relationships and cardinalities.
- Design a suitable ER diagram.
- Convert the ER model into a relational schema.
- Identify appropriate primary keys and foreign keys.

2. Database Implementation and Data Management

Implement the proposed database using SQL.

The implementation should demonstrate:

- Appropriate database structure creation and modification.
- Suitable integrity constraints.
- Insertion of realistic customer, policy, premium, claim, and payment data.
- Modification and deletion of records.
- Enforcement of data consistency and referential integrity.

The database should contain sufficient interrelated data to support meaningful analysis.

3. SQL-Based Data Processing and Analysis

Develop SQL queries to generate meaningful information from the database.

The queries should address scenarios such as:

- Customer policy portfolio.
- Policy-wise premium collection.
- Agent workload.
- Claim frequency.
- Claim approval rate.
- Claim settlement amount.
- Policy renewal statistics.
- Highest and lowest value claims.
- Customers with repeated claims.
- Policy categories requiring attention.

Appropriate techniques for data filtering, transformation, calculation, grouping, and summarization must be demonstrated.

4. Advanced SQL: Joins, Subqueries and Views

Develop advanced SQL solutions based on the requirements of the case study.

The implementation must include:

- Appropriate joins involving multiple relations.

- Subqueries for solving complex insurance-management requirements.
- At least one correlated subquery.
- At least two meaningful views for presenting derived insurance information.

The student must develop queries such as determining:

- Claims exceeding the average claim amount of their policy category.
- Agents handling above-average claim volumes.
- Customers with repeated claims.
- Policies having above-average premium revenue.
- Customers with multiple policies.
- Policies approaching renewal.
- Policy-level performance.
- Other relevant management-level insights.

The choice and purpose of each advanced query must be explained and justified.

5. Insurance Claim Investigation Risk Identification

Create an Insurance Claim Investigation Risk Identification and Decision-Support Module using the database designed by you. The insurance organization wants to identify claims that may require detailed investigation based on claim amount, previous claim history, policy age, claim frequency, and policy category.

Develop an appropriate solution that analyses claim patterns in relation to relevant benchmarks and generates a report such as:

Claim	Policy Type	Claim Amount	Previous Claims	Policy Age	Risk Status
Claim A	Motor	₹8,50,000	5	8 months	HIGH INVESTIGATION RISK
Claim B	Health	₹45,000	0	4 years	NORMAL

The criteria for classifying a claim as HIGH INVESTIGATION RISK, MODERATE RISK, or NORMAL must be defined and justified by the student.

The solution must integrate information from multiple relations and demonstrate effective use of advanced SQL and database-driven decision support.

CASE STUDY 6: University Hostel and Accommodation Management System

A university plans to develop a Hostel and Student Accommodation Management System to replace its existing manual and spreadsheet-based hostel management process. The proposed system should manage information related to students, hostels, rooms, beds, room allocations, hostel fees, attendance, maintenance complaints, visitors, and disciplinary records. It should support hostel allocation, room management, fee tracking, attendance monitoring, complaint management, and generation of analytical reports for university administration. The system must ensure data consistency, integrity, reduced redundancy, accurate room allocation, and efficient retrieval of hostel information. Analyse the given case study and independently design, develop, and evaluate a relational database solution for the proposed system. The entities, attributes, relationships, table names, and relational schema must be identified and designed by the student based on the requirements of the case study.

1. Requirement Analysis and Database Design

Analyse the case study and:

- Identify the functional requirements and business rules.
- Identify appropriate entities and attributes.
- Identify relationships and cardinalities.
- Design a suitable ER diagram.
- Convert the ER model into a relational schema.
- Identify appropriate primary keys and foreign keys.

2. Database Implementation and Data Management

Implement the proposed database using SQL.

The implementation should demonstrate:

- Appropriate database structure creation and modification.
- Suitable integrity constraints.
- Insertion of realistic student, hostel, room, allocation, fee, and complaint data.
- Modification and deletion of records.
- Enforcement of data consistency and referential integrity.
- Appropriate management of room and bed availability.

The database should contain sufficient interrelated data to support meaningful analysis.

3. SQL-Based Data Processing and Analysis

Develop SQL queries to generate meaningful information from the database.

The queries should address scenarios such as:

- Hostel occupancy.
- Room utilization.
- Student attendance.
- Hostel fee collection.
- Outstanding hostel fees.
- Maintenance complaints.
- Hostel-wise student statistics.
- Frequently reported maintenance problems.
- Disciplinary incidents.
- Students requiring administrative attention.

Appropriate techniques for data filtering, transformation, calculation, grouping, and summarization must be demonstrated.

4. Advanced SQL: Joins, Subqueries and Views

Develop advanced SQL solutions based on the requirements of the case study.

The implementation must include:

- Appropriate joins involving multiple relations.
- Subqueries for solving complex hostel-management requirements.
- At least one correlated subquery.
- At least two meaningful views for presenting derived hostel information.

The student must develop queries such as determining:

- Students having outstanding fees above the average outstanding amount.
- Hostels having occupancy above their capacity benchmark.
- Students with repeated maintenance complaints.
- Students with attendance below hostel average.
- Hostels having higher-than-average maintenance expenditure.
- Students with repeated disciplinary incidents.
- Hostel-level performance.
- Other relevant administrative insights.

The choice and purpose of each advanced query must be explained and justified.

5. Student Hostel Support Risk Identification

Create a Student Hostel Support Risk Identification and Decision-Support Module using the database designed by you. The university wants to identify students who may require administrative intervention based on attendance, outstanding hostel fees, maintenance complaints, and disciplinary history.

Develop an appropriate solution that analyses student hostel information in relation to relevant administrative benchmarks and generates a report such as:

Student	Hostel	Attendance %	Outstanding Fee	Complaints	Disciplinary Incidents	Risk Status
Student A	H01	62	₹18,500	4	2	HIGH PRIORITY
Student B	H03	91	₹0	1	0	NORMAL

The criteria for classifying a student as **HIGH PRIORITY**, **NEEDS ATTENTION**, or **NORMAL** must be defined and justified by the student.

The solution must integrate information from multiple relations and demonstrate effective use of advanced SQL and database-driven decision support.

CASE STUDY 7: Hospital Emergency Department Management System

A large metropolitan hospital wants to develop an Emergency Department Management System to improve the management of patients arriving at the emergency department. Currently, patient registration, doctor allocation, diagnostic requests, treatment records, and discharge information are maintained using separate systems. The proposed system should capture patient arrival, emergency priority, attending physician, department, diagnosis, laboratory investigations, treatments, medications, admission decisions, and discharge information. The hospital management wants the database to support both operational processing and emergency-care analytics.

Design and implement a relational database solution for the given scenario.

A. Conceptual and Logical Design

Independently determine:

- Functional requirements.
- Business rules.
- Entities and attributes.
- Relationships and cardinalities.
- ER diagram.
- Relational schema.
- Primary and foreign keys.

B. SQL Implementation

Implement the database using SQL and demonstrate:

- DDL commands.
- Constraints.
- DML operations.
- Realistic sample data.
- UPDATE and DELETE operations.
- Referential integrity.
- Appropriate date and numerical validations.

C. Analytical SQL

Develop queries to determine:

- Average emergency waiting time.
- Doctor-wise patient load.
- Priority-wise patient distribution.
- Department-wise emergency cases.
- Most frequent diagnoses.
- Average treatment duration.
- Patients requiring admission.
- Peak emergency hours.
- Doctors handling unusually high workloads.

D. Advanced SQL

Demonstrate:

- Multi-table joins.
- Nested subqueries.
- Correlated subquery.
- At least two views.

Develop queries to identify patients whose treatment duration exceeds the average for their diagnosis and doctors whose workload exceeds the department average.

E. Emergency Priority Decision System

Develop a Patient Emergency Priority Assessment Module.

Patient	Emergency Level	Waiting Time	Diagnosis Severity	Treatment Complexity	Priority Status
P101	Critical	42 min	High	High	IMMEDIATE
P205	Moderate	18 min	Medium	Low	STANDARD

Define and justify criteria for:

- IMMEDIATE
- URGENT
- STANDARD

CASE STUDY 8: University Placement and Recruitment Management System

A university wants to develop a centralized Campus Placement and Recruitment Management System. The system should maintain information about students, academic programs, companies, job profiles, eligibility criteria, placement drives, applications, aptitude tests, interviews, offers, and joining status. The placement office wants to analyse student employability, company participation, recruitment success, and placement performance.

Required Tasks

1. Database Design

Design independently:

- ER model.
- Relational schema.
- Keys.
- Constraints.
- Business rules.

2. Implementation

Implement the database and demonstrate:

- DDL.
- Constraints.
- Sample data.
- DML.
- Modification and deletion.
- Referential integrity.

3. SQL Analysis

Develop queries for:

- Eligible students for each company.
- Company-wise applications.
- Placement-drive participation.
- Selection rate.
- Department-wise placement percentage.
- Average package.
- Highest and lowest packages.
- Students receiving multiple offers.
- Companies with highest recruitment success.

4. Advanced SQL

Develop:

- Multi-table joins.
- Subqueries.
- Correlated subqueries.
- Two or more views.

Determine:

- Students selected at a salary above their department average.
- Companies with selection rates above the overall average.
- Departments performing below university placement average.

5. Student Placement Readiness Classification

Develop a Placement Readiness Decision-Support System.

Student	Department	CGPA	Aptitude Score	Interview Success	Offers	Readiness
S101	CSE	6.4	48	0	0	NEEDS INTERVENTION
S205	ECE	8.6	86	2	1	PLACEMENT READY

The student must define a suitable classification model using database-derived indicators.

CASE STUDY 9: Public Bus Transport Management System

A city transport corporation operates hundreds of buses across multiple routes. It wants to replace its fragmented records with a Public Bus Transport Management System. The system should manage buses, drivers, routes, stops, schedules, passengers, tickets, trips, maintenance, fuel consumption, and breakdowns. Management requires analytical information about route utilization, vehicle performance, and operational efficiency.

Required Tasks

1. Design Challenge

Independently identify:

- Entities.
- Attributes.
- Relationships.
- Cardinalities.
- Business rules.
- ER diagram.

2. Implementation Challenge

Implement:

- Bus records.
- Route information.
- Driver allocation.
- Trip schedules.
- Ticket transactions.
- Fuel records.
- Maintenance.
- Breakdown records.

3. SQL Analytics

Generate queries for:

- Route-wise passenger volume.
- Bus utilization.
- Driver workload.
- Ticket revenue.
- Fuel consumption.
- Breakdown frequency.
- Maintenance expenditure.
- Most profitable routes.
- Least utilized routes.

4. Advanced SQL

Demonstrate:

- Multi-table joins.
- Nested queries.
- Correlated queries.
- Views.

Determine:

- Routes with passenger volume above average.
- Buses with above-average maintenance expenditure.
- Drivers with unusually high trip workloads.

5. Route Efficiency Classification

Route	Passengers/Day	Revenue	Fuel Cost	Trips	Efficiency
R101	4,850	₹92,000	₹24,000	38	HIGHLY EFFICIENT
R205	1,250	₹25,000	₹23,500	31	LOW EFFICIENCY

Students must formulate an appropriate efficiency model.

Case Study 10: University Library Management and Resource Utilization System

A university library wants to replace its manual book circulation system with an integrated database. The system should manage books, authors, publishers, categories, copies, students, faculty members, borrowing transactions, returns, reservations, fines, and library branches. The system should provide analytical information about resource utilization and identify users who may require intervention because of repeated overdue returns.

1. Requirement Analysis and Database Design

Students must:

- Identify functional requirements.
- Identify entities and attributes.
- Determine relationships and cardinalities.
- Design an ER diagram.
- Identify PKs and FKs.
- Establish appropriate integrity constraints.

2. Database Implementation and Data Management

Implement:

- Book and author tables.
- User tables.
- Book-copy management.
- Borrowing and return transactions.
- Reservation records.
- Fine management.
- Constraints and referential integrity.
- Realistic sample data.

3. SQL-Based Data Processing and Analysis

Generate queries for:

- Most borrowed books.
- Most active users.
- Overdue books.
- Fine collection.
- Category-wise circulation.
- Branch-wise utilization.
- Books never borrowed.
- Popular authors.
- Monthly borrowing trends.
- Users with repeated overdue returns.

4. Advanced SQL: Joins, Subqueries and Views

Develop:

- Multi-relation joins.
- Subqueries comparing borrowing frequency with average borrowing.
- Correlated subqueries identifying users whose borrowing exceeds their peer group.
- View for book circulation statistics.
- View for user borrowing and fine history.

5. Library Usage Risk Identification

Create a Library Usage Risk Module.

Classify users based on:

- Number of overdue books.
- Outstanding fines.

- Repeated late returns.
- Lost or damaged books.

Generate:

User	Category	Overdue Books	Outstanding Fine	Risk Status
User A	Student	4	₹850	HIGH RISK
User B	Faculty	0	₹0	NORMAL

Case Study 11: Airline Reservation and Flight Operations Management System

An international airline plans to develop an Airline Reservation and Flight Operations Management System to replace its existing fragmented reservation and spreadsheet-based operational record management process. The proposed system should manage information related to passengers, airports, aircraft, airlines, routes, flights, flight schedules, seats, reservations, payments, baggage, cancellations, and flight operations. It should support passenger reservation, seat allocation, ticket management, baggage tracking, flight scheduling, and generation of analytical reports for airline management. The system must ensure data consistency, integrity, reduced redundancy, accurate reservation processing, and efficient retrieval of passenger and flight information. Analyse the given case study and independently design, develop, and evaluate a relational database solution for the proposed system. The entities, attributes, relationships, table names, and relational schema must be identified and designed by the student based on the requirements of the case study.

1. Requirement Analysis and Database Design

Analyse the case study and:

- Identify the functional requirements and business rules.
- Identify appropriate entities and attributes.
- Identify relationships and cardinalities.
- Design a suitable ER diagram.
- Convert the ER model into a relational schema.
- Identify appropriate primary keys and foreign keys.

2. Database Implementation and Data Management

Implement the proposed database using SQL.

The implementation should demonstrate:

- Appropriate database structure creation and modification.
- Suitable integrity constraints.
- Insertion of realistic passenger, flight, reservation, and payment data.
- Modification and deletion of records.
- Enforcement of data consistency and referential integrity.
- Appropriate handling of cancelled and completed reservations.

The database should contain sufficient interrelated data to support meaningful analysis.

3. SQL-Based Data Processing and Analysis

Develop SQL queries to generate meaningful information from the database.

The queries should address scenarios such as:

- Passenger travel history.
- Flight occupancy.
- Route-wise passenger volume.
- Aircraft utilization.
- Airport-wise flight statistics.
- Highest and lowest utilized flights.
- Revenue generated by routes.
- Cancellation statistics.
- Baggage statistics.
- Flights requiring operational attention.

Appropriate techniques for filtering, transformation, calculation, grouping, and summarization must be demonstrated.

4. Advanced SQL: Joins, Subqueries and Views

Develop advanced SQL solutions based on the requirements of the case study.

The implementation must include:

- Appropriate joins involving multiple relations.
- Subqueries for complex reservation and flight analysis.
- At least one correlated subquery.
- At least two meaningful views for derived airline information.

The student must develop queries such as determining:

- Flights having occupancy above the average occupancy of their route.
- Passengers whose travel frequency is above the overall passenger average.
- Routes having higher-than-average revenue.
- Aircraft with below-average utilization.
- Passengers with repeated cancellations.
- Airport-level flight performance.
- Other relevant airline management insights.

The choice and purpose of each advanced query must be explained and justified.

5. Flight Demand and Operational Risk Identification

Create a Flight Demand and Operational Risk Identification Module using the database designed by you. The airline wants to identify flights requiring operational attention based on occupancy, cancellation history, route demand, and aircraft utilization.

Develop an appropriate solution that analyses flight performance in relation to relevant benchmarks and generates a report such as:

Flight Route	Occupancy %	Cancellation Rate	Aircraft Utilization	Risk Status
AI201 Chennai–Delhi	94.5	8.2	91.4	HIGH DEMAND
AI305 Delhi–Mumbai	58.3	3.1	62.7	NORMAL

The criteria for classifying a flight as HIGH DEMAND, OPERATIONAL RISK, or NORMAL must be defined and justified by the student.

The solution must integrate information from multiple relations and demonstrate effective use of advanced SQL and database-driven decision support.

Case Study 12: Inventory and Warehouse Management System

A multinational manufacturing organization plans to develop an Inventory and Warehouse Management System to replace its spreadsheet-based inventory tracking process. The system should manage products, categories, suppliers, warehouses, inventory, purchase orders, purchase items, goods receipts, stock transfers, dispatches, and stock adjustments. It should support real-time inventory monitoring, procurement analysis, warehouse utilization, stock movement tracking, and management reporting.

1. Requirement Analysis and Database Design

Analyse the case study and:

- Identify functional requirements and business rules.
- Identify entities and attributes.
- Identify relationships and cardinalities.
- Design an ER diagram.
- Convert the ER model into a relational schema.
- Identify primary and foreign keys.

2. Database Implementation and Data Management

Implement:

- Product information.
- Supplier information.
- Warehouse information.
- Inventory records.
- Purchase orders.
- Stock receipts.
- Stock transfers.
- Dispatch records.

Demonstrate:

- DDL.
- DML.
- Constraints.
- UPDATE and DELETE operations.
- Referential integrity.

3. SQL-Based Data Processing and Analysis

Develop queries for:

- Warehouse-wise stock.
- Product-wise inventory.
- Supplier performance.
- Stock movement.
- Purchase expenditure.
- Fast-moving products.
- Slow-moving products.
- Products below reorder level.
- Warehouse utilization.
- Excess inventory.

4. Advanced SQL: Joins, Subqueries and Views

Develop queries to determine:

- Products below average inventory.
- Suppliers delivering above-average quantities.
- Warehouses having higher-than-average stock value.

- Products with repeated stock shortages.
- Correlated queries comparing product inventory with category averages.

Create:

- Inventory Status View.
- Supplier Performance View.

5. Inventory Shortage Risk Identification

Generate an inventory decision-support report:

Product	Warehouse	Current Stock	Reorder Level	Monthly Demand	Risk Status
P101	WH01	35	100	240	CRITICAL
P205	WH02	380	150	120	ADEQUATE

Students must define criteria for CRITICAL, REORDER REQUIRED, ADEQUATE, and EXCESS.

Case Study 13: Human Resource and Employee Performance Management System

A multinational organization plans to develop an Employee and Human Resource Management System to replace fragmented spreadsheet-based employee records. The system should manage employees, departments, designations, projects, attendance, leave, salary, performance evaluations, training programs, and promotions. The system should support employee administration, performance evaluation, workload analysis, attendance monitoring, and management reporting.

1. Requirement Analysis and Database Design

Students must independently:

- Identify functional requirements.
- Identify business rules.
- Identify entities and attributes.
- Determine relationships and cardinalities.
- Design an ER diagram.
- Develop the relational schema.
- Identify keys.

2. Database Implementation and Data Management

Implement:

- Employees.
- Departments.
- Designations.
- Projects.
- Attendance.
- Leave.
- Payroll.
- Performance.
- Training.

Demonstrate appropriate constraints, DDL, DML and referential integrity.

3. SQL-Based Data Processing and Analysis

Develop queries for:

- Department-wise employee count.
- Employee attendance.
- Leave utilization.
- Salary statistics.
- Project workload.
- Employee performance.
- Training participation.
- Promotion history.
- Department productivity.
- Employees requiring performance improvement.

4. Advanced SQL: Joins, Subqueries and Views

Develop:

- Employee-performance joins.
- Employees earning above departmental average.
- Employees whose performance is below department average.
- Employees with repeated absenteeism.
- Correlated queries comparing employees within departments.

Create:

- Employee Performance View.
- Department Workforce View.

5. BTL-6 Challenge: Employee Attrition Risk Identification

Generate:

Employee	Department	Performance Score	Leave Frequency	Promotion Gap	Risk Status
E101	IT	58.5	14	4 yrs	HIGH RISK
E205	HR	84.2	5	1 yr	NORMAL

Students must define and justify the criteria for HIGH RISK, MODERATE RISK, and NORMAL.

Case Study 14: Automobile Service and Maintenance Management System

An automobile service organization plans to develop an Automobile Service and Maintenance Management System. The system should manage customers, vehicles, vehicle models, service appointments, mechanics, repair jobs, spare parts, invoices, payments, and maintenance history. The system should support service scheduling, repair tracking, spare-part management, customer history analysis, and service-center performance analysis.

1. Requirement Analysis and Database Design

Students must identify:

- Functional requirements.
- Business rules.
- Entities and attributes.
- Relationships and cardinalities.
- ER diagram.
- Relational schema.
- Keys.

2. Database Implementation and Data Management

Implement customer, vehicle, service, mechanic, spare-part and billing information.

Demonstrate:

- DDL.
- Constraints.
- DML.
- UPDATE.
- DELETE.
- Referential integrity.

3. SQL-Based Data Processing and Analysis

Develop queries for:

- Customer service history.
- Vehicle service frequency.
- Mechanic workload.
- Spare-part usage.
- Service revenue.
- Most common repairs.
- Vehicle model-wise service statistics.
- Repeat service requirements.
- Maintenance expenditure.

4. Advanced SQL: Joins, Subqueries and Views

Develop queries for:

- Vehicles requiring more service than their model average.
- Mechanics handling above-average service jobs.
- Customers spending above-average maintenance costs.
- Frequently replaced components.
- Correlated vehicle-maintenance analysis.

Create:

- Vehicle Maintenance View.
- Mechanic Performance View.

5. Vehicle Maintenance Risk Identification

Vehicle	Model	Days Since Service	Repair Frequency	Maintenance Cost	Risk Status
V101	Model X	220	5	₹38,500	IMMEDIATE SERVICE
V205	Model Y	60	1	₹6,500	NORMAL

Criteria for risk classification must be independently defined.